Implementation of Clustering and Classification in Stock Asset Selection for Portfolio Diversification

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***Abstract*— This study introduces a novel machine learning framework for dynamic stock portfolio diversification, combining clustering, predictive modeling, and data-driven optimization to address challenges in emerging market volatility. Using comprehensive data from the Indonesia Stock Exchange enriched with technical indicators including Exponential Moving Average (EMA), Relative Strength Index (RSI), On-Balance Volume (OBV), beta coefficients, Moving Average Convergence Divergence (MACD), and macroeconomic inflation indicators, the framework systematically applies three unsupervised clustering algorithms: K-Means, DBSCAN, and Agglomerative Clustering to categorize stocks quarterly based on their return–risk profiles and behavioral patterns. Representative stocks from each identified cluster are strategically selected using a sophisticated multivariate Long Short-Term Memory (LSTM) neural network model specifically designed to forecast future Sharpe Ratios with enhanced accuracy. Market regimes including upward trending, downward trending, and sideways consolidation phases are separately classified using a robust Support Vector Machine (SVM) trained on carefully calibrated beta coefficient intervals. Portfolio weights are intelligently determined through an innovative data-driven allocation scheme that optimally balances the proportion of predicted Sharpe Ratios with classical mean-variance optimization principles across selected stocks. Comprehensive experimental results demonstrate that K-Means clustering consistently delivers the most balanced and diversified portfolios with the highest risk-adjusted Sharpe Ratios, while regime-aware optimization significantly improves overall portfolio performance during challenging sideways market conditions. The primary novelty lies in seamlessly integrating unsupervised clustering techniques, LSTM-based Sharpe Ratio forecasting capabilities, and intelligent data-driven weight allocation mechanisms into a unified comprehensive framework, enabling truly adaptive and responsive portfolio construction specifically tailored for volatile emerging markets.**

Keywords*— *stock portfolio, clustering, lstm, svm, market regime, portfolio optimization, sharpe ratio

I. Introduction

Investors in emerging markets encounter increased difficulties in constructing portfolios due to volatility and nonlinear market dynamics, which traditional correlation-based strategies frequently overlook [1]. Recent research has improved predictive performance by combining clustering with advanced deep learning models such as Long Short-Term Memory (LSTM), Recurrent Neural Network (RNN), and Gated Recurrent Unit (GRU) [2], including hybrid K-Means–LSTM frameworks that are optimized for multistep forecasting [3]. LSTM-based portfolio models have also demonstrated potential across different market regimes [4]. Nevertheless, the integrated application of clustering, time-series forecasting, regime classification, and regime-aware optimization is still relatively unexamined, particularly in emerging markets. Density-based clustering has been effective in uncovering intricate market structures [5], while machine learning methodologies continue to enhance the precision of portfolio optimization [6].

This study introduces a comprehensive framework that integrates quarterly clustering of Indonesian stock data, LSTM-based Sharpe Ratio forecasting, and Support Vector Machine (SVM) based market regime classification using beta values. These components are incorporated into a regime-informed mean–variance optimization process, resulting in an adaptive and data-driven approach to portfolio construction.

II. Methodology

A. Data Acquisition and Feature Selection

This research employs daily stock prices from the Indonesia Stock Exchange (IDX) covering the period from 2015 to 2025. There are 110 stocks were selected from 11 distinct sectors: Finance (e.g., BBCA.JK, BBRI.JK), Energy (e.g., ADRO.JK, BYAN.JK), Consumer Goods (e.g., KLBF.JK, ICBP.JK), Technology (e.g., EMTK.JK, MTDL.JK), Healthcare (e.g., SILO.JK, MIKA.JK), in addition to other sectors such as Property, Transportation, Basic Materials, and Infrastructure.

Six key features were identified based on established financial theories: daily returns, Sharpe ratio, Exponential Moving Average (EMA), Relative Strength Index (RSI), On-Balance Volume (OBV), and beta [7], [8]. Daily returns indicate short-term price momentum, while the Sharpe ratio provides a measure of risk-adjusted performance [9]. EMA represents trend-following behavior, RSI indicates momentum reversals [10]. OBV incorporates volume analysis to evaluate the sustainability of price movements, and the Beta coefficient assesses systematic risk in relation to the market [11]. This combination guarantees comprehensive coverage across the aspects of momentum, trend, volume dynamics, risk-adjusted performance, and systematic risk.

B. Clustering Process

Three clustering algorithms were selected based on complementary strengths:

1) *K-Means*: Selected for its computational efficiency and the ability to form compact, spherical clusters that are appropriate for financial assets exhibiting similar risk-return profiles. Its centroid-based methodology is in accordance with the mean-variance relationships of modern portfolio theory [12].

2) *DBSCAN*: Chosen as a density-based technique that can identify outliers and manage non-spherical clusters, which is crucial for recognizing stocks with distinct risk characteristics [13].
3) *Agglomerative Clustering*: Utilized for its hierarchical methodology employing Ward's linkage criterion, which minimizes within-cluster variance in alignment with the goals of portfolio optimization [14].

To classify stocks based on risk and performance attributes, six standardized financial metrics were combined into multidimensional vectors and normalized using z-score standardization. Temporal features were extracted using tsfresh for time-series embedding [15].

The elbow method and Silhouette score are used to determine the optimal number of clusters by analyzing the point where the reduction in within-cluster sum of squares (WCSS) begins to level off, ensuring a balance between model simplicity and clustering performance.

- K-Means: Optimal cluster number ($k = 8$) determined using Elbow Method (Fig. 1)
- DBSCAN: Parameters ($\varepsilon = 11.849$, min samples=9) optimized via k-distance plots and silhouette analysis (Silhouette Score=0.5375) (Fig. 3)
- Agglomerative: Ward's linkage also supported k=8 (Fig. 2)

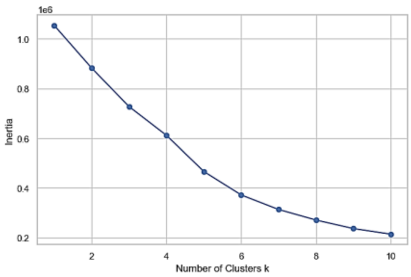

Fig. 1. Elbow curve showing optimal K-Means cluster count.

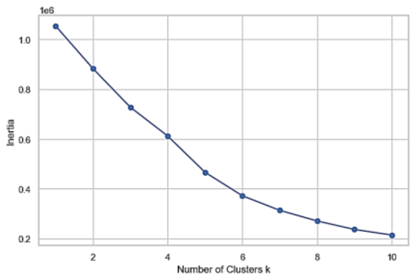

Fig. 2. Elbow curve showing optimal Agglomerative cluster count.

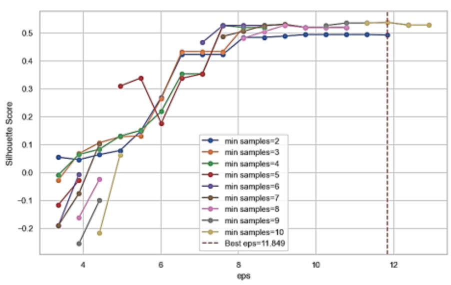

Fig. 3. Silhouette score for DBSCAN clustering.

C. *Forecasting Using LSTM*

The design of LSTM was purposefully developed to forecast financial time series: A two-layer LSTM architecture (comprising 50 units in each layer): This configuration achieves a balance between addressing the complexities associated with temporal dependencies and reducing the likelihood of overfitting [16]. The input features employed include: Moving Average Convergence Divergence (MACD), a technical momentum indicator, and inflation, a macroeconomic factor that affects asset valuation [17]. Training configuration: 20 epochs utilizing the Adam optimizer and Mean Squared Error (MSE) loss function, with validation performed through learning curve analysis [18]. An LSTM model is created for each representative cluster, utilizing historical data with a two-step input method to predict the third step, trained on data from Q4 2014 to Q4 2024.

D. *Portfolio Evaluation Metrics*

1) *Representative Stock Selection*

Representatives for each cluster are determined by LSTM predictive accuracy using Mean Absolute Error (MAE). Lower MAE indicates higher forecast accuracy and temporal consistency [19].

2) *Portfolio Weighting*

Weights are derived from predicted Sharpe Ratios using two-stage normalization:

- Min-Max normalization to a [0,1] range.
- Normalized to ensure the total allocation sums to one:

$$W_i = \frac{w_i^{\text{scaled}}}{\sum_j w_j^{\text{scaled}}} \tag{1}$$

This method ensures non-negativity, proportionality, and responsiveness to expected performance.

3) *Performance Metrics*

The performance metrics in this study are evaluated using three distinct methods, namely:

- Realized Portfolio Return

$$R_{p,t} = \sum_{i=1}^{n} w_{i,t} \cdot r_{i,t} \tag{2}$$

- Portfolio Risk and Variance

$$\sigma_p^2 = w^T \Sigma w,\ \ \sigma_p = \sqrt{\sigma_p^2} \tag{3}$$

- Sharpe Ratio of the Portfolio

$$SR_p = \sum_{i=1}^{n} w_i \cdot \frac{R_i - R_f}{\sigma_i} \tag{4}$$

E. *Regime-Aware Classification and Portfolio Optimization*

In order to align portfolio strategies with the changing dynamics of the market, this research integrates beta-based regime classification, SVM modeling, and regime-specific portfolio optimization utilizing the Mean-Variance Optimization (MVO) method.

1) Beta-Based Market Regime Classification

The SVM employing the RBF kernel was selected for classifying market regimes because of its robust performance in high-dimensional settings and its ability to adeptly handle non-linear relationships between beta values and market regimes [20]. The beta coefficient (5) measures sensitivity of market changes, classifying stocks into three separate regimes:

- Uptrend (Class 1): $\beta > 1 + \varepsilon$
- Sideways (Class 0): $\beta < \varepsilon$
- Downtrend (Class -1): $\beta < -1 - \varepsilon$.

$$\beta_t = \frac{\mathrm{Cov}(r_{i,t}, r_{m,t})}{\mathrm{Var}(r_{m,t})} \tag{5}$$

2) Regime-Specific Portfolio Optimization

After regime classification, portfolio optimization is conducted separately for each regime class $(-1, 0, 1)$ using the Mean-Variance Optimization (MVO) method. The optimization aims to minimize portfolio variance under full investment and no short-selling constraints, ensuring non-negative weights that sum to one. Once the optimal weights $w*$ are determined, key performance metrics, including portfolio risk (σ_p), expected return $(E[R_p])$, and Sharpe ratio (SR_p) are computed. This regime-based approach enhances adaptability to changing market conditions, with the resulting regime-specific weights serving as proportional indicators for strategic capital allocation.

III. Results

A. *Clustering Performance and Comparison*

The visualization of cluster distribution corroborates the quantitative findings of the study regarding the characteristics of each clustering method as seen in Fig. 4. K-Means generates 8 distinct, well-separated, and balanced spherical clusters, which supports the construction of a diversified portfolio. Similarly, agglomerative clustering results in 8 clusters, albeit with irregular shapes and varying densities, which reflects its hierarchical and risk-aware approach. In contrast, DBSCAN results in only 2 primary clusters, with 30–40% of the points classified as noise, thereby validating concerns regarding limited asset representation and inadequate diversification. Although all methods function within the same feature space, their partitioning philosophies diverge, K-Means allocates all points, Agglomerative constructs nested structures, and DBSCAN identifies outliers. These visual insights emphasize that K-Means is optimal for systematic diversification, Agglomerative is appropriate for risk-aware strategies, and DBSCAN, despite its high-quality clustering in limited areas, is not suitable for practical portfolio applications.

B. *Representative Stock Selection*

In the wake of the LSTM-based forecasting of Sharpe Ratios obtained from each clustering technique, a representative stock was chosen from each cluster to exemplify the most precise prediction performance within that group. The selection process utilized MAE metric, identifying the stock with the lowest MAE in each cluster as the representative. This methodology guarantees that the selected stock most accurately reflects the predictive reliability of the model within its designated cluster. Table I displays the representative stocks chosen for each clustering method, along with their associated MAE values and sector classifications.

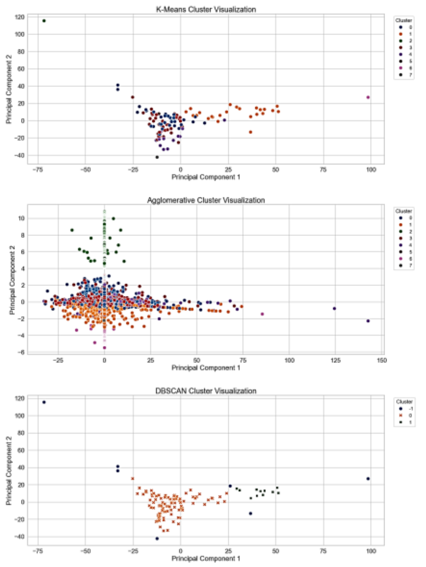

Fig. 4. Visualization of Clustering Results for K-Means (top), Agglomerative Clustering (middle), and DBSCAN (bottom).

TABLE I.
REPRESENTATIVE STOCKS SELECTED PER CLUSTERING METHOD

Method	Cluster	Ticker	MAE	Sector
K-means	0	MCAS.JK	0.625653	Tech
	1	BUMI.JK	0.531449	Energy
	2	CMPP.JK	0.473049	Transport
	3	PTSN.JK	0.407508	Tech
	4	MIKA.JK	0.294381	Health
	5	ATIC.JK	0.622857	Tech
	6	GGRM.JK	0.844251	Consumer non-cyclicals
	7	IMAS.JK	0.649052	Consumer cyclicals
Algomeratif	0	KBLI.JK	0.617522	Industry
	1	BBKP.JK	0.558085	Finance
	2	GGRM.JK	0.851559	Consumer non-cyclicals
DBSCAN	3	MIKA.JK	0.297464	Health
	4	PTSN.JK	0.387655	Tech
	5	ASGR.JK	0.789501	Finance
	6	BUMI.JK	0.580739	Energy
	7	SCPI.JK	0.758287	Health
	-1	BUMI.JK	0.544848	Energy
	0	MIKA.JK	0.296308	Health
	1	BBKP.JK	0.553753	Finance

The analysis of representative stocks based on MAE indicates a strategic trade-off between predictability and diversification. As can be seen in TABLE I, MIKA.JK stands out across all methodologies with the lowest MAE (0.294–0.297), which reflects its predictable and defensive characteristics. In contrast, GGRM.JK consistently underperforms (MAE >0.84) yet still receives significant allocation in K-Means, underscoring a tension between prediction accuracy and diversification. Sector analysis reveals varying levels of predictability—healthcare is bipolar, technology shows moderate consistency (with PTSN.JK as a notable exception), and consumer sectors exhibit high unpredictability. Among the methods evaluated, DBSCAN demonstrates the best average MAE but lacks diversity, K-Means provides balanced MAE with extensive sector exposure, and Agglomerative presents the worst MAE while offering the most comprehensive coverage. These results emphasize the necessity of balancing predictability with diversification, indicating that sector-specific predictability should inform clustering and stock selection.

C. *Integrated Portfolio Evaluation: Allocation, Performance Metrics, and Expected Returns*

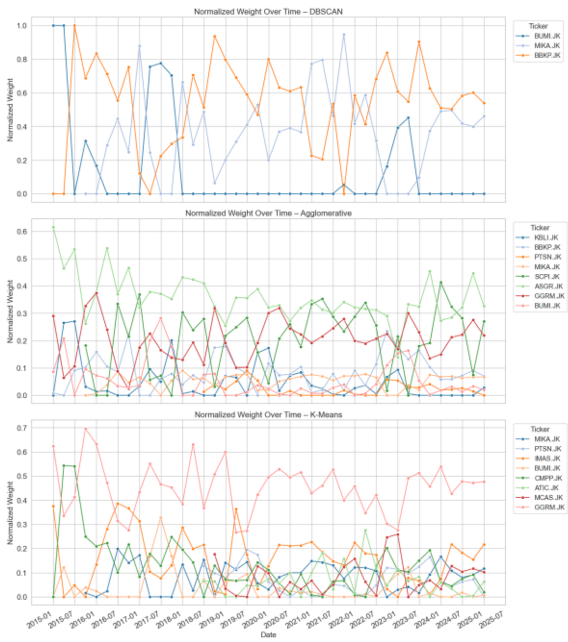

Fig. 5. Time-Series Portfolio Weight Allocation for DBSCAN (top), Agglomerative (middle), and K-Means (bottom).

Fig. 5 shows the portfolio weights allocation over time. The weight distribution visualizations highlight distinct behaviors across clustering methods. Agglomerative Clustering shows an erratic yet risk-focused allocation pattern, with early volatility (e.g., KBLI.JK at 61%) followed by dynamic sector rotation (SCPI.JK and ASGR.JK dominating 2018–2020), and stabilization during 2021–2025 with evenly distributed weights (0.1–0.3 per stock). This supports its suitability for risk-averse investors, offering balanced and low-risk portfolios—unlike the more return-driven K-Means. In contrast, DBSCAN exhibits highly concentrated and unstable allocations, dominated by only three stocks (MIKA.JK, BUMI.JK, BBKP.JK) with extreme binary switching and no sustained diversification. Its sensitivity to outliers and failure to maintain balanced allocations make it unsuitable for practical portfolio management, aligning more with speculative strategies.

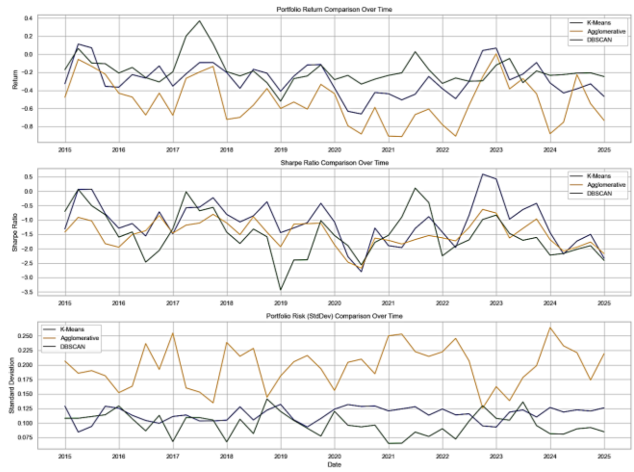

Fig. 6. Time-series visualization of Portfolio Return (top), Sharpe Ratio (middle), and Risk (bottom) for DBSCAN, K-Means, and Agglomerative Clustering methods.

Fig. 6 illustrates the time-series performance of portfolio metrics: return, Sharpe ratio, and risk, across DBSCAN, K-Means, and Agglomerative Clustering. The results reveal distinct behavioral patterns for each method. DBSCAN produces the highest return volatility (−0.4 to +0.4), reflecting unstable portfolio allocation, while Agglomerative demonstrates strong resilience and recovery potential, and K-Means delivers balanced, controlled performance. Sharpe ratio trends confirm K-Means as the most consistent (around −0.2), Agglomerative as occasionally positive, and DBSCAN as highly erratic (−0.6 to +0.4), highlighting its poor risk-adjusted performance. Risk levels form a clear hierarchy: DBSCAN highest, K-Means moderate, and Agglomerative lowest, supporting the interpretation that method choice significantly influences portfolio stability. Although returns tend to converge during 2021–2025, persistent differences in risk profiles indicate structural impacts beyond short-term performance. Overall, Agglomerative excels in risk control, K-Means supports balanced strategies, while DBSCAN remains unsuitable due to excessive volatility and weak Sharpe ratio.

D. *Market Regime-Aware Portfolio Strategy: Classification and Optimization*

Based on the classification evaluation using SVM on the labels generated by each clustering method, from showed result in TABLE II, it was found that the DBSCAN method delivered the best performance, with an average cross-validated F1-macro score of 0.639 and a test set accuracy of 0.67, despite producing only two main classes. The Agglomerative Clustering method showed moderate performance, achieving an average F1-macro score of 0.494 and accuracy of 0.56, indicating that its hierarchical structure provided better class separability than K-Means. Meanwhile, the K-Means method yielded the lowest classification performance, with an average F1-macro score of 0.486 and test accuracy of 0.49. These results highlight that the choice of clustering method significantly affects the quality of classification in stock asset selection for portfolio diversification, with DBSCAN excelling in identifying extreme patterns, while Agglomerative offers more structurally stable grouping.

TABLE II.
COMPARISON TABLE OF PORTFOLIO OPTIMIZATION RESULTS

Clustering Methode	Number of classes	F1-Macro (CV)	Accuracy (Test Set)	F1-Macro (Test Set)	Representative Ticker per Class
K-Means	3	0.486	0.49	0.42	-1: CMPP.JK, 0: ATIC.JK, 1: BUMI.JK
Agglomerative	3	0.494	0.56	0.53	-1: GGRM.JK, 0: SCPI.JK, 1: BBKP.JK
DBSCAN	2	0.639	0.67	0.67	-1: BUMI.JK, 1: MIKA.JK

The portfolio optimization results show clear differences across the three clustering methods. K-Means produced the most balanced regime allocations, with Class 1 (BBNI.JK) achieving the best performance (Sharpe Ratio: 0.2523), while other classes showed suboptimal returns. Agglomerative Clustering, in contrast, resulted in negative Sharpe Ratios across all classes, with Class 0 (SCPI.JK) exhibiting the worst outcome (Sharpe Ratio: -0.8940), indicating poor risk-return trade-offs. On the other hand, DBSCAN offered a mixed outcome: Class -1 (BUMI.JK) outperformed all others with a Sharpe Ratio of 0.4789, while Class 0 had no valid representative, and Class 1 (MIKA.JK) showed weak performance. These findings highlight that DBSCAN may provide more favorable portfolio regimes under certain conditions, whereas Agglomerative Clustering appears less effective for diversification-driven asset selection.

IV. Discussion

A. Performance and Economic Insights

The empirical findings indicate that various clustering methods produce significantly different portfolio performances within the Indonesian stock market. K-Means emerges as the most balanced method, consistently providing stable Sharpe ratios with moderate volatility, owing to its adherence to mean–variance optimization principles and its capacity to create well-distributed clusters [21]. In contrast, Agglomerative clustering leads to low-risk yet conservative portfolios, making it appropriate for risk-averse investors, albeit potentially sacrificing higher returns.

DBSCAN, on the other hand, demonstrates poor performance regarding portfolio diversification, as it tends to create highly concentrated clusters and categorizes large segments of data as noise. Nevertheless, its advantage lies in regime classification, which presents potential uses in tactical asset allocation strategies. The incorporation of LSTM forecasting underscores that healthcare stocks such as MIKA.JK show greater predictability (low MAE), while cyclical sectors prove more challenging to forecast due to their sensitivity to economic changes. This amalgamation of techniques offers prospects for data-driven, adaptive portfolio strategies.

B. Limitations and Practical Considerations

Several limitations must be considered when interpreting these findings. The study period (2015–2025) reflects specific market conditions in Indonesia, an emerging market with higher volatility and structural shifts, potentially limiting the generalizability of results to developed markets. The selected technical indicators (EMA, RSI, OBV, beta) and the MACD-inflation inputs for the LSTM model, though theoretically grounded, represent only one of many possible feature sets; different variables might produce varying clustering outcomes and forecast accuracy. The quarterly rebalancing schedule, while practical, may not suit all investor profiles or market scenarios. Moreover, transaction costs, omitted in the optimization framework, could erode gains due to frequent rebalancing, especially in a market with liquidity constraints affecting smaller stocks. Lastly, the regime classification based solely on beta values in an SVM model may oversimplify market complexity, omitting key dynamics such as volatility shifts, inter-asset correlations, or macroeconomic signals, and the use of only three regimes (uptrend, sideways, downtrend) might fail to capture the full market spectrum.

C. Future Research Directions

This research paves the way for numerous exciting avenues in future studies. By broadening the feature set to encompass essential metrics, macroeconomic factors, and alternative data sources like sentiment indicators or ESG elements, the quality of clustering and predictive precision could be significantly enhanced. Utilizing sophisticated ensemble methods, such as integrating DBSCAN for outlier identification, K-Means for primary clustering, and Agglomerative for fine-tuning, might produce more reliable outcomes. Investigating dynamic clustering that adjusts to changing market conditions and incorporates time-dependent correlations could provide a more accurate representation of financial market non-stationarity. Assessing various rebalancing frequencies while considering transaction costs and market microstructure would improve real-world applicability. Furthermore, expanding the framework to include multi-asset portfolios, such as bonds or options, and performing cross-market validation across both emerging and developed markets could bolster the generalizability and comparative significance of the suggested methodology.

V. Conclusion

This research presents a comprehensive machine learning framework designed for dynamic stock portfolio construction within the Indonesian market. It integrates clustering techniques, LSTM forecasting, SVM-based regime classification, and regime-aware optimization. Among the various clustering methods, K-Means consistently yields superior risk-adjusted returns due to its ability to create balanced and well-separated clusters, making it the most effective choice for diversification. The use of LSTM to forecast Sharpe ratios significantly enhances asset selection, particularly in sectors with predictable trends such as healthcare and technology. Additionally, SVM-driven regime-aware optimization allows portfolios to adjust to changing market conditions, although DBSCAN's higher classification accuracy is counterbalanced by its lack of diversification. This framework not only promotes methodological innovation but also offers practical insights for managing portfolios in emerging markets by quantifying the trade-offs associated with different clustering methods and highlighting sector-specific predictability. Despite certain limitations, including a limited feature set, the exclusion of transaction costs, and a focus on specific markets, the framework establishes a robust foundation for future improvements through the incorporation of broader data sources and cross-market validation. In summary, the study illustrates that the integration of various machine learning techniques can significantly enhance portfolio strategies in volatile and structurally dynamic markets.

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